

Designing a PDA

$$\delta(q, a, b) = \{ (q', c), \dots \}$$

$q \in Q$, $a = \text{input char} \in \Sigma_\epsilon = (\Sigma \cup \epsilon)$

$i=j$

~~a~~~~a~~~~b~~~~b~~~~c~~~~c~~

$$L = \{ a^i b^j c^k \mid i=j \text{ or } i=k, \text{ with } i, j, k \geq 0 \}$$

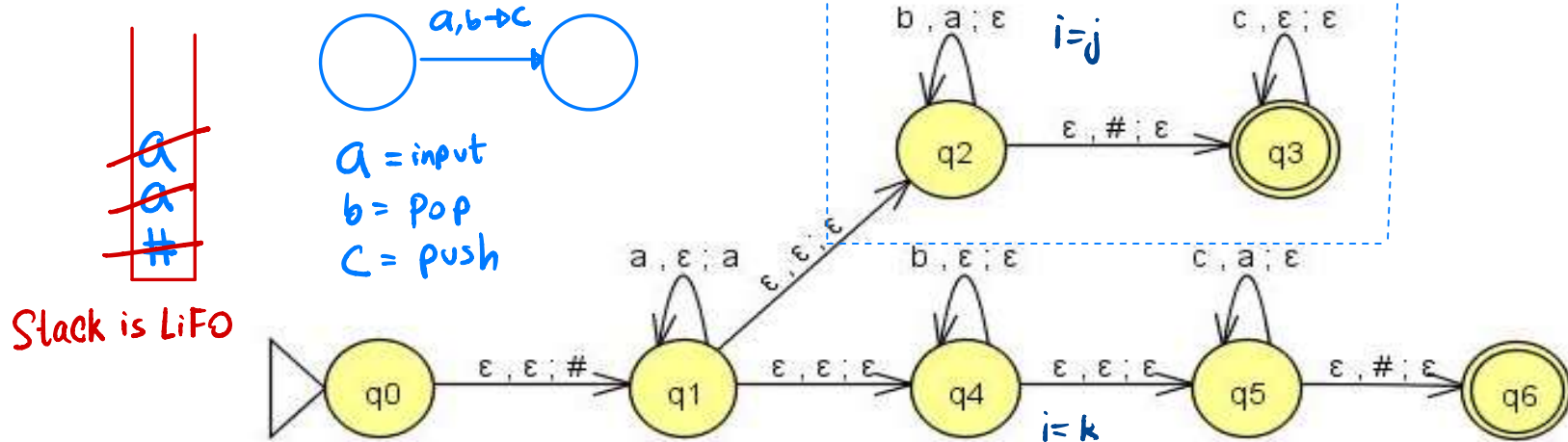
$\perp$

$b = \text{stack head} \in \Gamma_\epsilon$

$q' = \text{new state} \in Q$

$c = \text{push to stack} \in \Gamma_\epsilon$

Formal definition of PDA := NFA + Stack



Challenge construction

- Given word w , and character c , define

$\text{count}(w, c) = \text{count of } c \text{ in } w$

$w = abaa$

$\text{Count}(w, a) = 3$

- Consider the language

$$L = \{w_1 w_2 : w_1, w_2 \in \{0, 1\}^*, \text{count}(w_1, 0) = \text{count}(w_2, 1)\}$$

- Design a CFG and PDA for it

$w_1 \cdot w_2$

$w_1 = 00110$

$\text{Count}(w_1, 0) = 3$

$w_2 = 11001$

$\text{Count}(w_2, 1) = 3$

Challenge PDA construction

$$L = \{w_1 w_2 : w_1, w_2 \in \{0, 1\}^*, \text{count}(w_1, 0) = \text{count}(w_2, 1)\}$$

$w_1, w_2 = |00110|1100|$

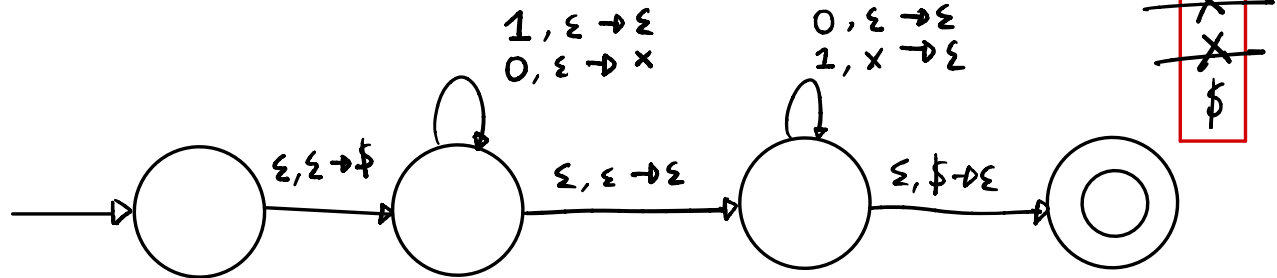
$w_1 = 00110$

$\text{Count}(w_1, 0) = 3$

$w_2 = 11001$

$\text{Count}(w_2, 1) = 3$

When reading 0 in w_1 , push x to stack.
When reading 1 from w_2 , pop x from stack.



Challenge CFG construction

$$L = \{w_1 w_2 : w_1, w_2 \in \{0, 1\}^*, \text{count}(w_1, 0) = \text{count}(w_2, 1)\}$$

Challenge: design a PDA

Build a PDA that accepts the language

$$\{x \in \{a, b\}^* : \#a \text{ in } x = \#b \text{ in } x\}$$

$$\#a = \text{Count}(n, a)$$

↓
Number of a's
amount of a's

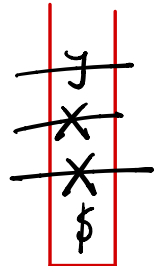
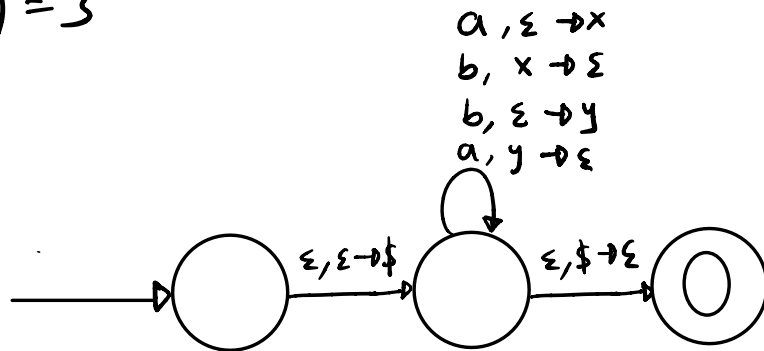
$x = ababba$ When a is read first push x to stack. Then when reading b pop x.
When b is read first push y to stack. Then when reading a pop y.

$$\text{Count}(x, a) = 3$$

$$\text{Count}(x, b) = 3$$

$$\Gamma = \{x, y, \$\}$$

$$\Sigma = \{a, b\}$$



Design a PDA

Build a PDA that accepts the language

$\{w \mid \text{the number of a's in } w \text{ is not same as the number of b's in } w\}$

$w = ababba$ } $bbb = \begin{bmatrix} y \\ y \\ y \\ \$ \end{bmatrix}$
#a=#b
Stack empty after
reading this part of
String

$\text{Count}(x, a) = 3$

$\text{Count}(x, b) = 6$

$w = ababba$ } $aa = \begin{bmatrix} x \\ x \\ \$ \end{bmatrix}$
#a=#b
Stack empty after
reading this part of
String

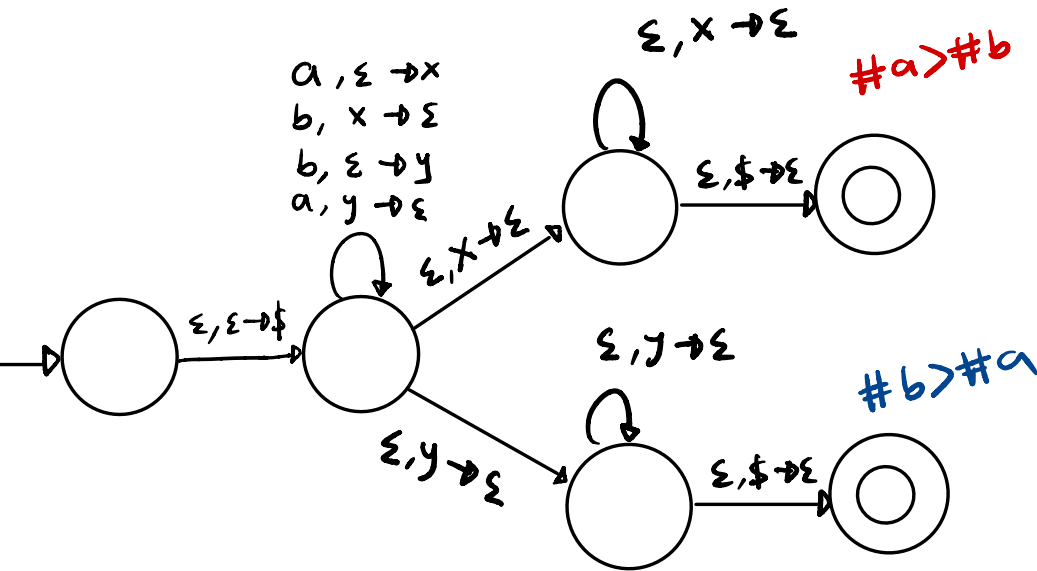
$\text{Count}(x, a) = 3$

$\text{Count}(x, b) = 4$

Design a PDA

Build a PDA that accepts the language

$\{w \mid \text{the number of a's in } w \text{ is not same as the number of b's in } w\}$



$w = ababba$

$\#a = 3$
 $\#b = 4$

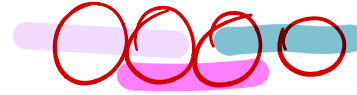
Stack empty after reading this part of String

$aa =$

1	x	1
x		
	\$	

$\text{Count}(x, a) = 3$
 $\text{Count}(x, b) = 4$

Design a PDA



Build a PDA that accepts the language

$\{x\#y \mid \text{the number of } 00 \text{ substring in } x \text{ is equal to the number of } 11 \text{ substring in } y\}$

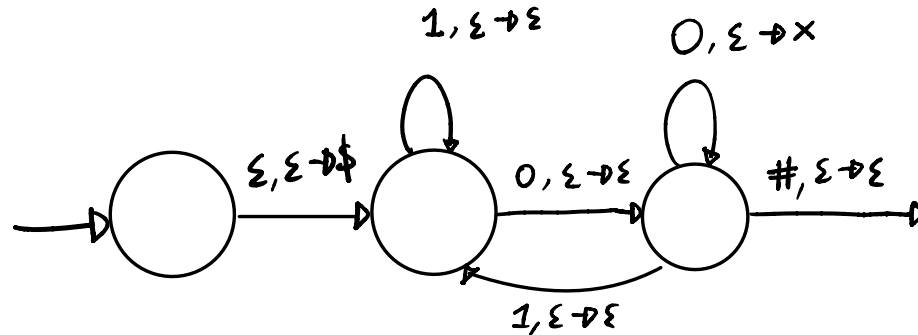
$1000100\#110111$
 $\underbrace{1000100}_x \quad \underbrace{110111}_y$

000 $\rightarrow$ 000
 000

$\text{Count}(x, 00) = 3$

$\text{Count}(y, 11) = 3$

X
X
X
\$



Design a PDA

$\underbrace{1000100}_x \# \underbrace{110111}_y$

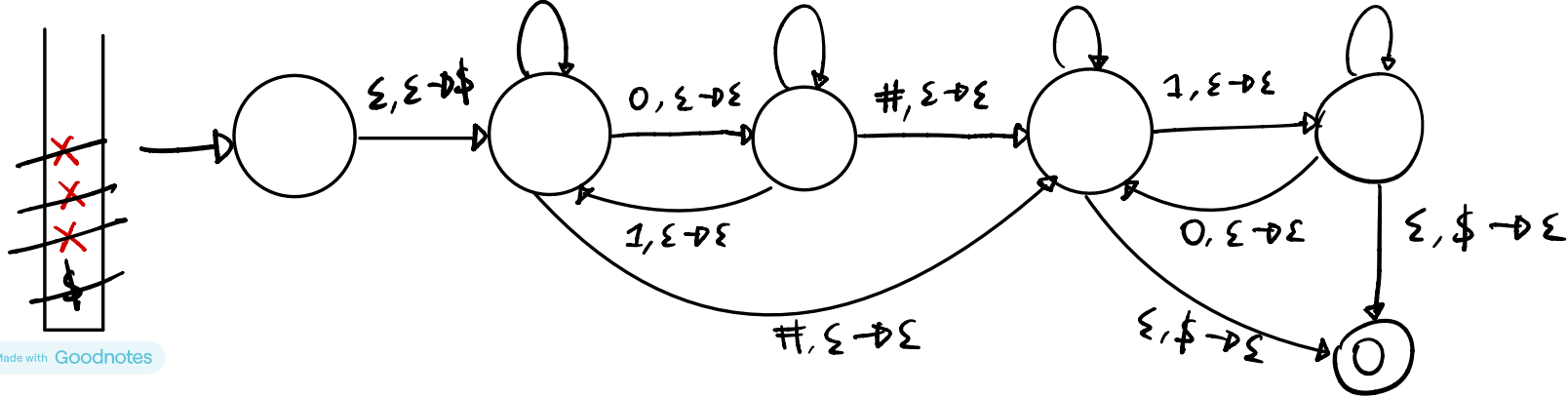
Build a PDA that accepts the language

$\{x\#y \mid \text{the number of } 00 \text{ substring in } x \text{ is equal to the number of } 11 \text{ substring in } y\}$

$\# \in L$

$\text{Count}(\#, 00) = 0$

$\text{Count}(\#, 11) = 0$



Design a PDA

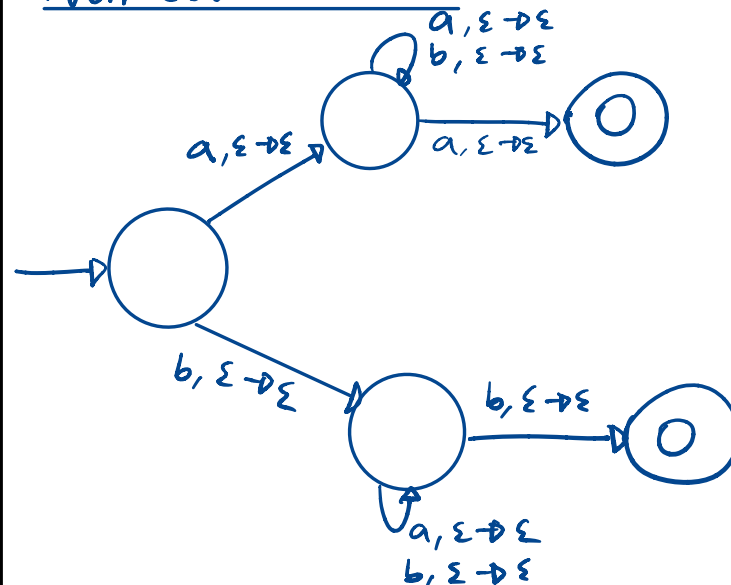
$$a(a+ b)^*a + b(a+ b)^*b$$

Build a PDA that accepts the language

$\{w \mid w \text{ starts and end with the same character}\}$

Deterministic PDA

Non-deterministic PDA



PDAs and CFGs are equivalently expressive

Theorem 2.20: A language is context-free if and only some nondeterministic PDA recognizes it.

Consequences

- Quick proof that every regular language is context free
- To prove closure of class of CFLs under a given operation, can choose two modes of proof (via CFGs or PDAs) depending on which is easier

~~*~~, ~~U~~, ~~Complement~~

$\{a,b\}^*$

aa, ab, ba, bb

$S \rightarrow AB$

Closure properties of CFLs

$G_1 = (V_1, T, P_1, S_1)$ and $G_2 = (V_2, T, P_2, S_2)$

Assumption: $V_1 \cap V_2 = \emptyset$, that is, non-terminals are not shared

$CFG \rightarrow CFL \rightarrow PDA$
recognizes it.

Theorem

CFLs are closed under union. L_1, L_2 CFLs implies $L_1 \cup L_2$ is a CFL.

Theorem

CFLs are closed under concatenation. L_1, L_2 CFLs implies $L_1 \cdot L_2$ is a CFL.

Theorem

CFLs are closed under Kleene star.

If L is a CFL $\Rightarrow L^*$ is a CFL.

$a^* = \{\epsilon, a, aa, \dots\}$

$(a+b)^*$ $S \rightarrow aS \mid bS \mid \epsilon$ $S \rightarrow aS$
 $\rightarrow aaaS$
 $\rightarrow aaaa$

$S \rightarrow aS \mid \epsilon$

More bad news: CFL not closed under intersection

Theorem

CFLs are *not* closed under intersection.

Theorem

CFLs are *not* closed under complement.

Example

$$L = \{ a^i b^j c^k \mid i=j \text{ or } i=k, \text{ with } i,j,k \geq 0 \}$$

To design a CFG that generates L ...

$$L = \{ a^i b^j c^k \mid i=j, \text{ with } i,j,k \geq 0 \} \cup \{ a^i b^j c^k \mid i=k, \text{ with } i,j,k \geq 0 \}$$

$$S \rightarrow Sc \mid T$$

$$T \rightarrow aTb \mid \varepsilon$$

Example

$$L = \{ a^i b^j c^k \mid i=j \text{ or } i=k, \text{ with } i,j,k \geq 0 \}$$

To design a CFG that generates L ...

$$L = \{ a^i b^j c^k \mid i=j, \text{ with } i,j,k \geq 0 \} \cup \{ a^i b^j c^k \mid i=k, \text{ with } i,j,k \geq 0 \}$$

$$S \rightarrow T \mid aSc$$

$$T \rightarrow Tb \mid \varepsilon$$

Designing a CFG – challenge!

Building a CFG to describe the language

$$\{ ww \mid w \in \Sigma^* \}$$

ww

$$V = \{ S \}$$

$$\Sigma = \{ a, b \}$$

$$R = ?$$

S

This language is not context free!

This can be proved using a
“pumping lemma” for context free languages

This lang. req. copying, not just counting.

ww^R

$w = abaa$

$w^R = aabaa$

$ww^R = abaaabaa$

$S \rightarrow aSa \mid bSb \mid \epsilon$